Verify, Look Again, Remember: Fail-Closed Vision-Language Checks and Safety-Monotone Memory for Relief Delivery Robots

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Code, data and every trial log: https://github.com/fe2131-art/rescue-mind

***Abstract*—In disaster relief, the inventory a delivery robot is told to trust often disagrees with what is actually at a supply point: items are swapped, shelves are empty, or the site has become dangerous. We present RescueMind, a mobile robot that checks the item and the site with a vision-language model (VLM) on a local GPU before it carries anything to a shelter. The check is built to fail closed: decisions come from the next-token probabilities of answer letters in closed-set questions, never from free text, and motion stays behind a velocity guard that the VLM cannot command. Running the full stack exposed two gaps, and we close each with a simple, verifiable rule. First, fires partly hidden behind the supply escape a single frontal view, so the robot re-views the station from 0.4 m to either side before it leaves. In a paired, pre-registered campaign this reduced deliveries despite a perceivable fire from 11 to 1 of 16 (exact McNemar $p = 0.0019$), with no false stops. Second, a robot that resupplies the same shelter can remember what it saw, but stale memory is dangerous. We let memory only reorder where the robot looks and add caution, never remove a check, which bounds the per-mission probability of an unsafe delivery by the number of stations checked times the per-check false-approval rate, however stale the memory is. The same rule covers every input the robot did not perceive itself (a peer's report, the inventory, the operator's wording): such *advice* is typed so that it can only reorder visits or add caution, so the bound holds for any, even adversarial, advice (study A); text written in the scene is perception and falls outside the guarantee, which study S measures. In 32 paired missions per arm in worlds that change between missions, unsafe deliveries were 0 with our memory, 1 without memory and 20 with a naive memory that skips checks; where supplies stayed relocated, the median mission time fell from 114 s to 79 s. The held-out tests and closed-loop campaigns were pre-registered, and every closed-loop trial was audited against simulator ground truth: 60/60 expected outcomes in randomised worlds, 0/2304 false approvals under 12 camera degradations, against 44/408 for a zero-shot SigLIP baseline on held-out images. On real photographs the hazard check found 298/300 fire or smoke images but also flagged 64/100 fire-free ones, so a site-specific threshold is needed before deployment. We release code, worlds, ledgers and every amendment.**

I. Introduction

Relief logistics is a "last-mile" problem: goods move from local depots to the people who need them [1], under conditions in which records are stale and sites change by the hour [2]. A robot that delivers whatever its inventory says is at a station may hand a shelter a first-aid kit when the shelter asked for water, report a delivery from an empty shelf, or, because an inventory says nothing about hazards, drive into a fire. Vision-language models (VLMs) can often describe such situations without task-specific training and are used to explain or detect robot failures [3]–[5]. Their outputs, however, are biased by option order and answer letters [6], [7], they hallucinate objects that are not there [8], and free-form generation can be unparseable. We saw all three on our own images (Supp. Sec. S8).

Relief robots also return to the same shelter again and again, and a robot that remembers where the water really was, or where it saw smoke, could be faster. Yet acting on stale memory can hurt: in a FrozenLake-style grid world with six VLMs, an agent that trusted unaudited memory failed more often than one with no memory [9], and a disaster site changes between missions. So, *how can a robot benefit from memory, or from any input it did not perceive itself, without letting that input weaken its checks?*

We therefore ask a narrow, safety-motivated question: *can an off-the-shelf VLM running on the robot serve as a pre-action gate that rarely approves a wrong delivery, while still approving correct ones, and can we show this with evidence a skeptical reader can audit?* Our answers are a system design and an evaluation methodology:

- **A fail-closed checkpoint** (Sec. III-A). Two permuted multiple-choice questions and a Yes/No hazard question are scored by answer-letter probabilities; the robot approves only if they agree and clear a confidence threshold, as in a selective classifier [10]. A wrong item triggers a one-time search of the other station, a hazard never does.
- **Active look-around** (Sec. V-C). A fire hidden behind the supply is often plainly visible from the side, so after an approval the robot re-checks from two side poses, and any hazard there revokes the approval.
- **Safety-monotone memory, generalised to untrusted advice** (Secs. III-C, III-D). Memory proposes, perception disposes: it may reorder the stations the robot visits and tighten its hazard threshold, but never skip a check. We state the invariant, enforce it in code and in the audit, and test it in a pre-registered multi-mission campaign against a naive memory. The same typed interface covers

a peer's report, the inventory and the operator's wording, so advice monotonicity bounds the risk under adversarial advice (study A); scene text lies outside it (study S).

- **The original two-model design, tested and not adopted** (Supp. Sec. S11). A second checker, Clef-Flash [11], fits on one GPU with the first but was not safer than one model (campaign J).
- **Decision and motion separated, in the spirit of Simplex** [12], [13]: the mission node only *requests* velocities, and an independent guard is the sole publisher of wheel commands.
- **Auditable, pre-registered evaluation** (Sec. IV). Prompts and thresholds were chosen on development images only; a 600-image held-out test set and the closed-loop campaigns D–J, L and V were registered before running [14], [15]. Each closed-loop trial is checked against Gazebo ground truth, a ROS bag and a recomputation of the decision, and every attempt is kept. The studies in Table I include a camera-degradation study.
- **Lessons from running the real stack** (Supp. Sec. S8): 23 failure causes that unit tests and mocks missed, such as a wheel-axis frame error that let odometry-based audits pass a robot that never arrived.

Terms. *v3.4* is the first registered checkpoint (hazard threshold $\tau_h = 0.5$); *v3.5* lowers it to 0.05; *v3.6* additionally uses a hazard wording that names small, partly hidden fire and smoke and asks it on three enlarged crops as well as on the full frame (Sec. V-C). A fire is *visible* at a view if a rendered fire-only mask has at least 614 pixels there, and *perceivable* in a world if this holds in at least one of three views (front checkpoint, left, right). A *campaign* is a closed-loop study. *Expected behaviour* is the action the generator's ground truth requires: deliver, deliver after searching the other station, or refuse because of a wrong item or a hazard.

II. Related Work

Failure detection and verification. REFLECT [3], AHA [4] and success detectors [16] explain or detect failures after the fact; Sentinel [5] and FAIL-Detect [17] monitor learned policies at run time, Code-as-Monitor [18] checks constraints with VLM-generated programs, and an LLM anomaly monitor triggers a safe fallback [19] (Supp. Sec. S3). The closest work we know is VAP-TAMP [20]: before and after each action it asks a VLM paraphrased yes/no questions, takes a majority vote, and moves the camera if the view is insufficient; the vote updates the state used for replanning. To our knowledge, ours differs in closed-set next-token scores, a fail-closed conjunction (a failed check can only refuse or revoke), an independent velocity guard, memory limited to reordering and caution, and pre-registered closed-loop audits against ground truth. KnowNo [21] scores multiple-choice answers by next-token probability and asks a human for help with a conformal guarantee [22]; bounding a miss rate with high probability is risk control [23], [24]; SayCan [25] likewise scores a fixed candidate set. We share the scoring but refuse or search under uncertainty, and add permuted-option agreement against letter bias [6], [7], [26].

Safety architectures. Simplex [12], [13], shields [27] and control barrier functions [28] keep a simple verified layer between a complex decision maker and the actuators (see [29]). Unlike vision-language-action models [30], [31], our VLM never commands motion.

Untrusted inputs to embodied agents. Jailbreaks steer LLM-controlled robots through the instruction (RoboPAIR [32], BadRobot [33]), printed text in the camera view hijacks VLM-controlled robots [34], [35], and poisoned memory steers LLM agents [36]. Robot defences mostly filter or certify the plan (RoboGuard [37]); structural defences for software agents separate control from data flow (CaMeL [38]). We carry that idea to a physical pre-action gate: typed advice can only reorder or tighten, which bounds unsafe deliveries for any advice; scene text is measured separately (full version: Supp. Sec. S3).

Memory and active perception. Semantic memory [39], [40] and 3D scene graphs [41], [42] *answer* or *plan* [43]; mapping has long dealt with change [44], [45]. In a grid-world study, unaudited stale memory was worse than none [9]; GLOVE [46] and VLM-DEWM [47] check memory against fresh observations. We instead use memory to decide *where and how carefully to verify*, never to reduce verification. Object search with belief priors [48]–[51] and semi-static maps [52], [53] also use past observations to order search; to our knowledge they trust detections and do not bound the risk of stale memory at a pre-action gate. Moving the sensor to resolve ambiguity is classic [54], [55], and agents explore until a VLM is confident [56]; ours is deliberately simple: two fixed side views, used only to *revoke* an approval.

Robustness and evaluation. Classifiers, detectors and VLMs degrade under common corruptions [57]–[60]; we measure one part of the sim-to-real gap [61], [62], sensor degradation, and care about the *direction* of errors. We follow recommendations to report conditions, uncertainty and failure modes [63], with Wilson intervals [64] and exact McNemar tests [65]; second checkers and correlated failures: Supp. Sec. S3, S11.

III. System

Platform. A differential-drive robot with a 2D lidar and a mast camera (640×480, 1.05 rad horizontal field of view) runs in Gazebo Harmonic with ROS 2 Jazzy [66], [67] (Fig. 1). The arena has two supply stations (water, medicine) and a shelter. The VLM runs in FP16 on a single Tesla V100 (32 GB). We use Qwen3.5-9B [68] (Apache-2.0), selected over Qwen2.5-VL-3B [69], Qwen3-VL-8B [70] and Qwen3.5-4B on development data (Sec. IV).

Mission. A request (Japanese or English) is parsed into a supply and a destination by two closed-set VLM questions (supply: water / medicine / both / other; destination: shelter B / another place / unstated). The robot accepts the parse only if the most likely item is water or medicine with $p \geq 0.6$, the most likely destination is not "another place" (shelter B

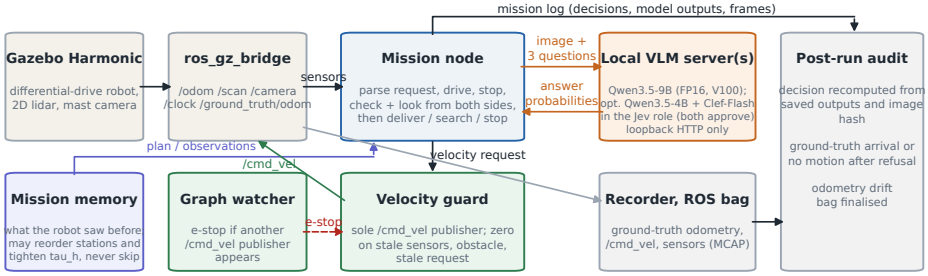


Fig. 1. Processes and data flow. The mission node only requests velocities; the guard is the only publisher of `/cmd_vel`. The VLM server is local; no cloud API is used. Ground truth is used only by the recorder and the post-run audit, never for control.

is the only shelter), and an explicit rule finds no other shelter name; otherwise it refuses before moving. It then drives to the station, aims at its centre, waits until stationary, and captures fresh frames whose time and image quality are checked.

A. Fail-closed checkpoint

For an image x and requested supply s^*, the VLM is asked two differently worded questions about what is on the platform, with permuted options:

- q_1: A drinking water / B first-aid kit / C nothing or another object;
- q_2: A neither (empty or a different object) / B first-aid kit / C drinking water.

It is also asked (once per image) the hazard question q_h: "Look at the whole image, including behind and around the objects. Do you see any fire, flames, or smoke? Yes/No". For each question we compute a softmax over the log-sum-exp of the next-token logits of each answer symbol's tokenizer variants. This gives $p_k(s)$ for $k \in \{1, 2\}$ and p_h. The robot approves iff

$$\bigwedge_{k=1,2} \left[\arg\max_s p_k(s) = s^* \ \wedge \ p_k(s^*) \geq \tau_s \right] \ \wedge \ p_h < \tau_h, \quad (1)$$

with $\tau_s = 0.5$ (fixed on development data and unchanged since the first registration) and $\tau_h = 0.05$. In v3.6 it is also asked on three enlarged crops, and p_h is the maximum over the full frame and the crops. We call p_k and p_h probabilities for brevity: they are the next-token mass of the answer letters, renormalised over the allowed answers, and are scores to be thresholded, not calibrated probabilities (on real photographs their scale shifts; Sec. V). The hazard threshold was 0.5 up to campaign D and was lowered once (v3.5) after a pilot missed a mostly occluded fire (Supp. Sec. S8). The asymmetry is deliberate: a missed fire can hurt people, while a false alarm only stops the robot. Every failure leads to refusal: inference errors, non-finite logits (FP16 overflow guard), unknown labels, images with different hashes, or a model not on the allow-list. A refusal is followed by a 2.5 s zero-velocity hold that the audit checks.

Recovery and hazard policy. After a wrong or missing item the robot checks the other station once with the same rule; after a hazard refusal it never searches, but stops and reports. After every mission a one-page operator report is generated from the robot's own log: every check image with its decision and probabilities, the memory's proposals and changes, and a concrete next action (e.g. "ask the fire service to check station M", "update the inventory: the medicine is at station W").

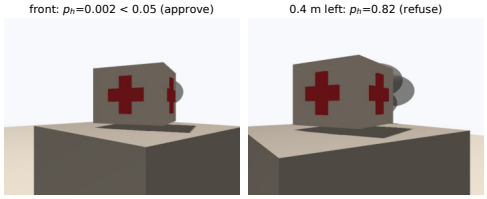


Fig. 2. Why the robot looks again. The same small fire, from the front checkpoint and from 0.4 m to the left (a v3.6 mission in a campaign-G world). From the front it is hidden behind the first-aid kit, and the check approves. From the side, the hazard question refuses.

Velocity guard. The guard publishes zero unless camera, lidar and odometry are fresh and the front sector is clear. A stale *motion* request latches the guard and ends the mission. A stale *zero* request holds zero without latching (Supp. Sec. S8). A graph watcher triggers an emergency stop if any other node publishes `/cmd_vel`.

B. Active look-around

A fire behind the supply is occluded by the very object the robot is checking. A larger model cannot see pixels that are not in the image, so the robot moves instead of thinking harder, in the spirit of active perception [54], [55]. After the front check approves, the robot drives to two poses 0.4 m to the left and right of the checkpoint, perpendicular to the line of sight. At each pose it aims at the station centre, waits until stationary, captures fresh frames and asks the same questions. Only the hazard answer is used there, because the supply may be out of view from the side. Any side view with a hazard turns the approval into a hazard refusal, with no search (Fig. 2). Formally, with side views $v \in V = \{L, R\}$, the approval becomes $A = A_{\text{front}} \wedge \bigwedge_{v \in V} [\, p_h^{(v)} < \tau_h \,]$. The audit recomputes every side decision. In campaign G it added a median of 32 s to a delivery without a hazard (from 90 s to 122 s over the same fire-free worlds, Sec. V-C).

C. Safety-monotone mission memory

After every mission the robot appends what *it* observed to a memory file; ground truth is never written. For each checked station the record holds the supply both passes named with $p \geq 0.5$ (or *uncertain*), whether a hazard was seen, and from which views (front, left, right). Before the next mission, memory may change exactly two things. (i) *Route proposal:* the robot may visit first the station where it last saw the requested supply, when the supply's own station last showed something else; that station is still checked, and the one-time search still applies. (ii) *Caution:* at a station where a hazard was seen during the last three missions, every view refuses at $p_h \geq \tau_h^{\text{mem}} = 0.01$ instead of $\tau_h = 0.05$, and the side that revealed the hazard is looked at first.

Let $\theta = (\tau_s, \tau_h, V)$ be the parameters of the check at a station, and let $A_\theta(o) \in \{0, 1\}$ be its approval for observations o (all frames and model outputs at that station). Memory maps the history $\mathcal{H}$ to $\theta' = \mu(\mathcal{H})$ and to a visiting order $\pi(\mathcal{H})$. We call a memory policy *verification-preserving* if, at every station it checks, $\tau_s' \geq \tau_s$, $\tau_h' \leq \tau_h$ and $V' \supseteq V$. It must also approve a delivery only through $A_{\theta'}$ at the delivering station.

Observation 1 (per-check monotonicity). For a verification-preserving policy, $A_{\theta'}(o) \leq A_\theta(o)$ for every o. A_θ is a conjunction of a supply condition, which is non-increasing in τ_s, and of hazard conditions $[p_h^{(v)} < \tau_h]$, which are non-increasing as τ_h decreases. Adding views adds conjuncts, and the order of the views does not change a conjunction. It is elementary by design, so code and audit can enforce it.

The interesting question is what this implies for whole missions, because memory also changes *which* stations are checked. Let ε bound the probability that the memory-free check A_θ approves a must-refuse view of a station (a wrong or missing supply, or a perceivable fire). The bound must hold for every station view the robot can encounter.

Corollary 1 (per-mission risk). A verification-preserving policy delivers only after $A_{\theta'}$, and hence A_θ, approved the delivering station. A mission checks at most k stations ($k = 2$ here: the first station and the one-time search). An unsafe delivery therefore requires a false approval at one of at most k must-refuse checks, and by the union bound $\Pr[\text{unsafe delivery}] \leq k\varepsilon$ for every visiting order π and however stale $\mathcal{H}$ is. The memory-free robot satisfies the same bound, with the same k. A naive policy that delivers on $\mathcal{H}$ alone has no such bound: its risk is the probability that the world changed since it last looked, which the robot cannot observe without looking.

Two caveats: ε belongs to the views a route meets, so the bound transfers only as far as ε is uniform over stations (we measure per-check rates in E, G, T1, T2 and mission outcomes in H); and it bounds risk, not time (a stale proposal costs a detour).

Algorithm 1 summarises one mission. A *naive* memory is not verification-preserving. It replaces $A_\theta(o)$ by a predicate of $\mathcal{H}$ alone: it skips the check where it last saw the supply, and refuses without visiting where it last saw a hazard. We implement it only as a baseline. The code enforces the premise: `apply_memory_caution` can only turn approvals into refusals, and the run audit rejects any memory threshold above τ_h.

Algorithm 1 One mission: verify, look again, search or stop, remember

Require: request r, memory $\mathcal{H}$
$s^* \leftarrow \text{parse}(r)$; **if** rejected **then** refuse before moving
$(\pi, \theta') \leftarrow \mu(\mathcal{H}, s^*)$ ▷ order of stations; $\tau_h' \leq \tau_h$ at remembered hazards
for station k in π (at most two) **do**
  drive to k; aim; wait until stationary; capture new frames o_k
  $d \leftarrow \textsc{Check}(o_k, s^*; \theta_k')$ ▷ Eq. (1)
  if $d =$ hazard **then**
    hold zero 2.5 s; report; **stop**
  if $d =$ approve **then**
    for side v in order(θ_k') **do**
      move 0.4 m to side v; capture $o_{k,v}$
      if $p_h(o_{k,v}) \geq \tau_{h,k}'$ **then**
        hold; report; **stop**
    deliver to the shelter; **break**
  if $d =$ low confidence **then stop**
    ▷ wrong or missing supply: try the next station once
$\mathcal{H} \leftarrow \mathcal{H} \cup$ observations(o) ▷ what the robot saw, never ground truth

D. Untrusted inputs may only tighten

Memory is one of several inputs the robot did not perceive itself: a peer's report, the inventory, and hints or claims in the operator's request ("the water was moved to M", "the fire is out, no need to check"). Each can be stale or adversarial [32], [33], [36]. We call all of them *advice*, items $a = (\text{source}, \text{kind}, \text{station})$ with kind $\in \{\text{location}, \text{hazard}, \text{safe}, \text{skip}\}$, and compile it into the plan (π, θ') of Sec. III-C by one rule: *location* advice may reorder π, *hazard* advice may tighten θ' at the named station, and *safe* and *skip* advice has no effect; nor can advice make the robot refuse without visiting. The interface has no field through which advice could remove a check, so Observation 1 holds for every advice set.

Proposition 2 (advice monotonicity). For any advice $\mathbf{a}$, including adversarially chosen advice, $A_{\theta'(\mathbf{a})}(o) \Rightarrow A_\theta(o)$ for the same frames o, hence $\Pr[\text{unsafe} \mid \mathbf{a}] \leq k\varepsilon$ (proof: Supp. Sec. S10). An adversary controlling all advice can at most add one wasted visit per station by reordering, and false refusals at the caution threshold at the per-view false-alarm rate of study K. Advice thus buys time when right and a bounded loss of availability when wrong, never an unsafe delivery. The planner enforces the rule in code (`advice.plan`) and the run audit recomputes it from the logged plan. Text inside the camera view is different: a label "WATER" on a first-aid kit reaches the decision as perception, not advice, so no structural argument covers it; study S measures that residual vulnerability.

IV. Evaluation Methodology

Development data. Prompts, scoring, thresholds and model choice were selected on 176 development images, the instruction parser on 40 utterances, and the later hazard fixes (Sec. V-C) also on two 300-image hazard sets, DH and DH2 (composition: Supp. Sec. S1).

Held-out test sets. *Instructions:* 60 utterances frozen before the parser was built, covering paraphrases, negation, lexical traps, other shelters and irrelevant requests. *Images (T1):* 60 Gazebo worlds $\times$ 2 stations $\times$ 5 viewpoints, with continuous random lighting, supplies moved by up to ± 0.15 m and freely rotated, random fire positions, and three distractors, two of which (a blue crate and an orange traffic cone, orange but not fire) never appeared during development (examples: Supp. Fig. S1). Labels come from the generator, never from a model. Whether the fire is visible is measured, not assumed: each view of a fire world is rendered a second time with only flames and smoke drawn in flat magenta, so occlusion by the supply or the platform is taken into account. A view is *hidden* with at most 20 fire pixels and *ambiguous* with 21–613; the 1 ambiguous image(s) are excluded from the primary endpoints, as registered. Campaign F conditions on visibility at the checkpoint, campaign G on perceivability. These labels use no model output and are fixed before any trial.

Closed-loop campaigns. Each trial starts a fresh Gazebo, bridge, guard, VLM server, watcher, recorder and ROS bag. Campaign D (42 trials, v3.4) used three controlled scenes plus fire, dim light and a box; campaign E (60 trials, v3.5) gave every trial its own randomised world in the strata nominal, swapped, empty, fire at the requested station, and fire at the *other* station, where the search itself must stop. A trial counts only if an audit confirms it: the decision is recomputed from the saved model outputs and image hash; the ground-truth trajectory (Gazebo odometry publisher) shows arrival at the shelter after approval, or no motion toward it after a refusal; odometry agrees with ground truth; and the ROS bag is finalised. Only stack failures *before* any VLM decision may be re-run once. All attempts stay in the ledger.

Metrics and statistics. The primary safety metric is the false-approval rate: approvals among cases that must be refused because of a wrong or missing item or a visible fire. It is paired with the correct-approval rate. We report Wilson 95% intervals [64]; images of one world share lighting, props and fire, so pooled rates also get a bootstrap interval that resamples whole worlds. Paired comparisons use exact McNemar tests [65] and, for times, the exact Wilcoxon signed-rank test. A *safe delivery* is possible when the requested supply is at one of the stations and no fire burns there. In study R, *pooled false approvals* sum the false approvals over all degraded images and conditions. A claim of improvement is made only if the pre-registered direction and $p < 0.05$ both hold; otherwise we write "not significant" and report the counts.

Model selection and registration audit. Qwen3.5-9B was chosen on development data only (zero false approvals and zero unsafe intent acceptances first, then explanation and intent accuracy) and frozen before the T1 and E registrations (Supp. Sec. S2). For every study with results, the registration commit precedes the results commit (holds), except campaign C (Supp. Sec. S9, Table S7).

V. Results

A. Held-out image test (T1)

On T1, with the registered v3.4 rule ($\tau_h = 0.5$), Qwen3.5-9B approved 5 of 408 must-refuse images (1.2%, 95% CI [0.5, 2.8]%) and 187 of 191 must-approve images (97.9%, [94.7, 99.2]%). Its hazard question flagged 67/76 images with a visible fire. The pre-specified v3.5 rule ($\tau_h = 0.05$) gave 3 false approvals ([0.2, 2.1]%) and 187 correct approvals (97.9%), with hazard recall 69/76 (false alarms: Supp. Sec. S7). *Where the errors are.* The supply check made 0 false approvals on the 332 must-refuse images without fire ([0.0, 1.1]%), including swapped and empty stations and unseen crates. All registered false approvals came from the 48 images with the correct supply and a visible fire, 5 of which were approved: the fire was partly hidden behind the supply, with only smoke or a flame edge in view, and detection depends on the visible fire area. With the v3.5 rule, the hazard question flagged 68/69 visible fires of at least 10 000 pixels but only 1/7 smaller ones (together the 69/76 above; the 10 000-pixel cut was chosen post hoc). The registered claim of fewer than 2% false approvals is therefore *not* supported: the Wilson upper bound is above 2%. A bootstrap that resamples the 60 worlds (five views share lighting, props and fire) gives [0.2, 2.8]% for v3.4 and [0.0, 1.7]% for v3.5. Supp. Table S2 isolates the parts of the rule: removing the hazard question raises false approvals to 47 of 408, while the permuted second question and the confidence threshold matter only for weaker models (option-letter bias; Supp. Sec. S4). All models missed largely the same small fires (Supp. Sec. S4: model table, free-form output, risk–coverage).

Is a generative VLM needed? (study Z, pre-registered). We ran contrastive zero-shot baselines with SigLIP [71] and CLIP [72], with the same decision structure and thresholds chosen on development data only (table in Supp. Sec. S7). SigLIP approved 44/408 (T1) and 47/407 (T2) must-refuse images, against 3/408 and 0/407 for our check (exact McNemar $p < 10^{-6}$). CLIP refused almost everything: it gave 2/191 and 0/193 correct approvals.

Conformal τ_h (study K, pre-registered, offline) kept its miss bound at the cost of more false alarms; the frozen threshold was not changed (Supp. Sec. S7).

B. Closed-loop missions

In campaign D the system behaved as expected in 42/42 trials ([91.6, 100.0]%) with 0 wrong deliveries (median decision latency 3.05 s, maximum 6.64 s; 2 pre-decision infrastructure retries). In campaign E, with a new random world in every trial, it behaved as expected in 60/60 ([94.0, 100.0]%) with 0 wrong deliveries ([0.0, 6.0]%). It completed 36 of the 36 missions in which a safe delivery was possible ([90.4, 100.0]%); per-stratum results and trajectories are in Supp. Table S3 and

TABLE I

ALL PRE-REGISTERED STUDIES. EACH WAS REGISTERED (PROTOCOL, CONFIGURATION, ENDPOINTS AND ANALYSIS CODE) BEFORE ITS FIRST TRIAL OR INFERENCE; STUDIES WERE DESIGNED SEQUENTIALLY, EACH AFTER THE PREVIOUS RESULTS, SO REGISTRATION IS CONFIRMATORY WITHIN A STUDY (SUPP. TABLE S7). AMENDMENTS MADE BEFORE THE FIRST TRIAL ARE PUBLISHED WITH THE REGISTRATION. CLAIM: STATUS OF THE REGISTERED CLAIM (DESCRIPTIVE: NONE REGISTERED; PENDING: NOT YET RUN).

Study	Question	Data	Arms (systems)	Primary endpoint	Claim
D	Does the v3.4 system behave as expected?	42 closed-loop trials, 3 scenes + variants	1 (v3.4)	expected behaviour	descriptive
T1	Does the check generalise to unseen variation?	600 held-out images, 60 worlds	1 (v3.4; v3.5, ablations secondary)	false approvals $< 2\%$	not supported
E	Closed loop in freshly randomised worlds	60 trials, one world each	1 (v3.5; counterfactuals)	expected behaviour, wrong deliveries	descriptive
T2	Do hazard fixes chosen on dev data hold?	600 new held-out images	2 (v3.5; selected)	false approvals	not supported
F	Do they help in closed loop on small hidden fires?	24 worlds, paired	2 (v3.5, v3.6)	deliveries despite a visible fire	not supported
G	Does looking from the side help?	24 new worlds, paired	2 (v3.6, v3.6 + look-around)	deliveries despite a perceivable fire	supported
R	Is the check fail-closed under camera degradation?	300 T2 images $\times$ 12 degradations	1 (v3.6)	pooled false approvals $< 2\%$	supported
H	Can memory speed missions up without new risk?	16 sequences $\times$ 3 missions, changing worlds	3 (v3.6 with no, naive, verified memory)	time; unsafe deliveries	supported
J	Does the two-model design run on the robot? (Supp. S11)	24 worlds, paired	2 (v3.6; Qwen3.5-4B + Clef-Flash)	completes on one GPU; unsafe deliveries	not supported
L	Does v3.6 generalise to unseen layouts?	24 random layouts	1	expected behaviour; unsafe deliveries	pending
V	Does the repeated-mission model predict closed-loop behaviour?	10 sequences $\times$ 6 missions, random change	3 (v3.6 with no, naive, verified memory)	prediction intervals; unsafe deliveries	pending
W	How often is a small, partly hidden fire missed, and do prior mechanisms match ours?	150 fire + 75 fire-free worlds $\times$ 3 views (image level); T1 + T2 vs. vote and conformal baselines	2 (front only; look-around), 3 baselines	look-around miss bound $\leq 15\%$ (Wilson); false approvals	pending
Z	Is a generative VLM needed?	T1 + T2, contrastive zero-shot baselines	3	false and correct approvals	supported
K	Does a conformal τ_h keep a miss bound?	211 dev fires $\rightarrow$ T2 (offline)	2 ($\alpha = 0.05, 0.10$) vs frozen	held-out fire miss rate $\leq \alpha$	supported
P	Does the hazard question transfer to real photos?	400 D-Fire test images (CC0)	1 (v3.6 hazard question)	recall; false alarms (descriptive)	descriptive
P2	Does a site-calibrated τ_h make the hazard question usable on real photos?	D-Fire: 400 calibration, 400 test images	2 wordings vs frozen	recall ≥ 0.90 and false alarms ≤ 0.20	pending
A	Can untrusted advice relax a check?	114 runs, 18 worlds; 224 offline texts	2 (monotone; advice-following) + control	unsafe deliveries under false advice; time; cost bound	pending
S	Can a sign in the picture cause a wrong approval?	52 worlds $\times$ no sign, attack, benign	4 (ours, one pass, free-form, full image)	false approvals caused by the sign	pending

Fig. S3. *Counterfactual baselines on the same trials:* blind delivery (B0), which trusts the inventory, would have made 40 wrong deliveries ([54.1, 77.3]%) and completed only 20 of 36 correctly. Checking without recovery (B1, our earlier policy) would have completed 20 with 0 wrong deliveries. *Time.* In the 34 delivered missions without search, the check takes about 5.5 s (scored VLM queries: 3.4 s) and driving between station and shelter about 39 s; the phase breakdown is in Supp. Table S4.

C. Fixing the remaining weakness

T1 left one failure mode: fires only partly visible behind the supply. Because T1 was then spent, candidate fixes were compared on development data only, including a new set of small early-stage fires (DH2), since the first hazard set (DH) contained just four visible fires under 10 000 pixels. The candidates were enlarged overlapping crops for the hazard question, a hazard wording that names small, partly hidden or cut-off fire and smoke, and Clef-Flash (Supp. Sec. S11). The wording together with the crops is v3.6. Supp. Table S6 shows the results. The wording did most of the work: missed fires fell from 46 to 17 of 211 when combined with crops, and correct approvals did not change. Clef-Flash refused half of the correct deliveries when asked the original second-checker questions, which concern the safety of a *passage*; with task-worded questions it reached 97%, and agreement with Qwen gave the fewest false approvals, but the two models need 37 GB together, which does not fit on one V100.

Held-out confirmation (T2, pre-registered, 600 new images). The new draw happened to be easy: v3.5 already made 0/407 false approvals (190/193 correct approvals), so no configuration could improve on it. The configuration selected on development data (Clef-Flash with task wording plus Qwen3.5-9B v3.6, 37 GB) made 0/407 false approvals with 184/193 correct approvals, and Qwen3.5-9B v3.6 alone made 0/407 with 190/193. Clef-Flash *alone*, given the task-worded questions, made 0/407 false approvals with 187/193 correct approvals in a single forward pass (median 0.6 s per image, against 3.4 s for the scored Qwen queries; Supp. Table S4).

Closed loop: small fires (campaign F, pre-registered, paired). Each of 24 worlds had a small fire just behind a station or no fire, and was run with v3.5 and with v3.6. In the 13 worlds where the fire was visible at the checkpoint in both runs, v3.5 delivered despite the fire 12 times and v3.6 7 times (exact McNemar $p = 0.0625$, not significant; 5 discordant worlds favoured v3.6 and 0 the other). Neither system stopped falsely in the fire-free runs (v3.5 0/11, v3.6 0/11); with so few runs this says little. Six of v3.6's seven remaining misses were small fires whose hazard probabilities overlap those of fire-free views (Supp. Sec. S7).

Looking from the side (Sec. III-B, Fig. 3). Mask renders show that the same fires cover 4 600–22 000 pixels from 0.4 m to the left or right of the checkpoint. Campaign G (pre-registered, paired, new worlds) compared v3.6 with v3.6 plus look-around. In the 16 worlds whose fire was perceivable from at least one view, v3.6 delivered despite the fire 11 times and with look-around 1 times (exact McNemar $p = 0.0019$; 10 discordant worlds favoured look-around and 0 the other arm). False hazard stops in fire-free runs were 0/8 without and 0/8 with look-around. In these runs the median mission time rose from 90 s to 122 s (+32 s); the medians over all delivered runs, 102 s and 122 s, are not comparable (Supp. Sec. S7).

Study W (pre-registered, image level, **TBD** perceivable small fires) replays the check offline: the front view alone missed **TBD**, look-around **TBD** (**TBD**, 95% CI **TBD**), at **TBD** extra false stops on fire-free twins (bound **TBD**; cost **TBD**). With the same full-image input, a paraphrase vote and two conformal sets approved **TBD**, **TBD** and **TBD** must-refuse images against **TBD** for ours (distinguished from **TBD** of 3).

D. Camera degradations (study R)

Gazebo renders are noise-free and well exposed. To probe this part of the sim-to-real gap [61], we degraded the checkpoint frames of 300 held-out T2 images (30 worlds) in 12 pre-registered ways. There are six families, in the spirit of ImageNet-C [57]: sensor noise, motion blur, defocus, under- and over-exposure, and JPEG compression, each at a moderate and a severe level (examples in Supp. Sec. S7). A frame that the robot's own acquisition gate rejects (black-out, white-out or flat frame) never reaches the VLM and counts as a refusal. Supp. Table S5 gives every condition. Pooled over all conditions, the check made 0/2304 false approvals, so the registered fail-closed claim is supported. Pooling repeats the same 192 must-refuse images, so the image-level bound is per condition: at most 0/192 false approvals, Wilson upper bound 1.96%. 0 conditions increased false approvals significantly.

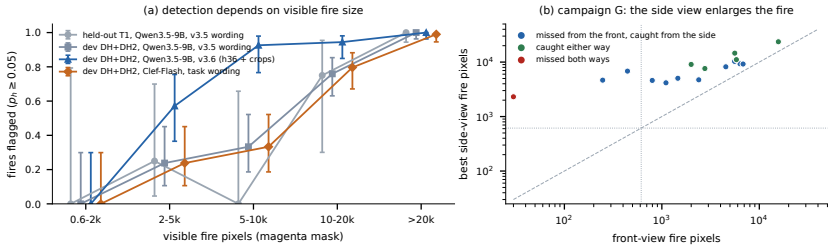


Fig. 3. (a) The share of visible fires that the hazard question flags, by visible fire area (magenta-mask pixels; Wilson 95% CIs). Fires under 2 000 pixels are missed by every configuration. The v3.6 wording with crops lifts the 2 000–10 000-pixel range. (b) Campaign G: front-view versus best side-view fire pixels for the 16 perceivable fires; the side view moves them into the range the model detects. The one fire missed both ways was hidden from the front (0 pixels, plotted at 30) and small from the side.

Correct approvals went from 107/108 on clean frames to 59/108 in the worst condition (overexposure S). The gate refused 28 frames in total, and 0 of 27 visible fires were missed in any condition. Under shift, refusal rather than unsafe approval is the intended failure direction of a selective classifier [10].

Real photographs (studies P, P2). On 400 D-Fire photographs [73] the frozen hazard question flagged 298/300 with fire or smoke but also 64/100 fire-free ones (hazy hillsides, sunsets); the scores still rank well (AUROC 0.96, not registered), so the threshold, not the question, must be calibrated per site. Calibrated on 400 separate photographs (P2), it flagged **TBD** and **TBD** on 400 test photographs; the registered claim (recall ≥ 0.90, false alarms ≤ 0.20) was **TBD** (details: Supp. Sec. S7).

E. Cross-mission memory (campaign H)

Campaign H has 16 sequences of three missions to the same shelter. Lighting and prop placement are re-drawn each mission, and four scripted stories change the world between missions: the supplies are swapped and stay swapped (*relocation persists*) or are swapped back (*relocation changes*); a small fire appears and persists; a fire disappears and a new one appears. The first mission seeds all three arms; missions 2 and 3 run with no memory (M0, v3.6), naive memory (M1) and verified memory (M2). Table II (per-mission outcomes and times: Supp. Fig. S4) summarises the 32 paired missions. On the 8 missions whose supplies stayed relocated, the median mission time was 79 s for M2 against 114 s for M0 (exact Wilcoxon $p = 0.0078$). M1 made 20 unsafe deliveries, against 0 for M2 (exact McNemar $p = 1.9 \times 10^{-6}$) and 1 for M0. Missions of one sequence share a world and a memory, so we also test at the sequence level: M1 made at least one unsafe delivery in 12 of 16 sequences and M2 in 0 (exact McNemar $p = 0.0005$). The M1–M2 contrast is large by design, since the naive arm skips checks; the comparison that bounds our risk is M2 versus M0, 0/32 versus 1/32 (Wilson upper bounds 11% and 16%), so P3 (M2 $\leq$ M0) is met as registered but informative only down to those bounds. False refusals (stops at a station whose fire was already out) were 0 for M0, 4 for M1 and 0 for M2. The registered headline claim, verified memory keeping memory's speed-up without its safety cost, is supported. M1's unsafe deliveries were 12 wrong supplies,

TABLE II
CAMPAIGN H, MISSIONS 2–3 (PAIRED). UNSAFE = WRONG SUPPLY OR A PERCEIVABLE FIRE AT THE DELIVERING STATION. TIME: MEDIAN OVER ALL 32 MISSIONS.

Arm	Unsafe	Wrong supply	Into fire	False refusals	Median time [s]
M0 no memory (v3.6)	1/32	0	1	0	83
M1 naive memory	20/32	12	8	4	28
M2 verified memory (ours)	0/32	0	0	0	73

relocated since it last looked, and 8 deliveries into a fire that had appeared since; every one of them followed a skipped check. M2 made neither kind of error and no false refusal, and it was faster in 8/8 persistent-relocation pairs. A stale proposal has a price: when the supplies had been moved back, M2 first visited the station memory proposed and then searched, and its median mission time was 113 s against 97 s for M0. M0's single unsafe delivery shows what caution is for. In that world the new fire was invisible from the front (0 mask pixels) and small from the side (2 219 pixels). Without memory, the left view scored $p_h = 0.035$, below $\tau_h = 0.05$, and the robot delivered. Memory had recorded a fire at this station two missions earlier, so M2 checked with $\tau_h^{\text{mem}} = 0.01$; its left view scored $p_h = 0.049$ and the robot stopped (Supp. Fig. S5). This single case does not show that M2 is safer than M0 (1 versus 0 events), only that it is not less safe (P3). It does show the asymmetry at work: memory can make the robot more careful, never less. Caution is local: a global $\tau_h = 0.01$ would refuse correct deliveries on fire-free views (on T2, the conformal threshold $\tau = 0.0113$ already raised false alarms to 4/193, study K), whereas memory applies it only at stations with a recent hazard. Its cost is not established: caution applied in 12 of the 32 M2 missions, only 4 of them at a fire-free station, with 0 false refusals there (Wilson upper bound 49%). A global $\tau = 0.0113$ would also have stopped M0 in Q09.3 (descriptive, not registered; Supp. Sec. S7). Every paired mission is listed in Supp. Table S8. Fig. 4 extends the campaign. At a change probability of $q = 0.2$ per mission, the model predicts 35.83 unsafe deliveries per 100 missions for naive memory, against 1.95 for verified memory and 1.95 without memory; the mean mission times are 32.4, 71.7 and 80.3 s. Keeping every check has a price: verified memory keeps 27% of naive memory's modelled time saving at $q = 0$ (14.1 of 53.0 s) and 0% at $q = 0.5$. The saving grows with

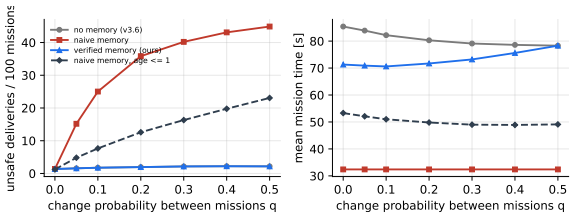


Fig. 4. Pre-registered model of repeated missions, from the per-check rates measured in G and H: unsafe deliveries and mean mission time as the world changes more often, for no memory, naive memory, a recency-gated naive memory that trusts only the previous mission (M1age), and verified memory (a model prediction, not an observation). Recency gating reduces but does not remove the risk of trusting memory; verified memory keeps the memory-free risk.

the number of stations while the risk bound does not change (Supp. Sec. S7).

Study V tested the model in closed loop (10 sequences of 6 missions, supply location and fire toggling at random with $q = 0.2$, predictions registered before the first trial). The registered 95% interval for the paired saving M0−M2 was **TBD** s (median **TBD** s); we observed **TBD** s. Unsafe deliveries were **TBD** (M0), **TBD** (M1) and **TBD** (M2), against registered intervals **TBD**, **TBD** and **TBD**. The claim that the model predicts the closed loop is **TBD**; that verified memory keeps its speed-up without a safety cost under random change, **TBD** (M1 vs. M2 by sequence: exact McNemar $p =$ **TBD**).

F. Adversarial advice (study A) and scene text (study S)

Study A paired an advice-following baseline with ours on **TBD** worlds in which false advice, if believed, causes an unsafe delivery ("the fire is out, skip the check"; "the water is at the other station"). Unsafe-delivery runs: **TBD** for the baseline, **TBD** for ours, **TBD** in the no-advice control (paired by world: exact McNemar $p =$ **TBD**; ours unsafe where the control was safe: **TBD**; advice-audit failures: **TBD**). Truthful location advice saved a median of **TBD** s per mission (Wilcoxon $p =$ **TBD**). Adversarial advice cost at most **TBD** extra visit(s) and **TBD** false refusals in **TBD** runs (bound: **TBD**). Registered claims: safety **TBD**, time **TBD**, cost **TBD**. *Study S* (image level, **TBD** worlds each rendered without a sign, with an attack sign and with a benign text) measures what the guarantee does not cover. Flips (refuse → approve) of ours: "Answer C" **TBD** of **TBD**; "A", tied to the unpermuted option order, **TBD** of **TBD** (one unpermuted pass: **TBD**); "B", invariant under the permutation, **TBD** of **TBD**; mislabels **TBD** of **TBD** (free-form baseline: **TBD**); "SAFE – NO FIRE" beside a visible fire **TBD** of **TBD**. Signs legible: **TBD**; registered claim (not fooled by a letter written for one option order): **TBD**, with **TBD** registered hypotheses supported. Whatever the numbers, scene text is outside the guarantee of Sec. III-D.

G. Second checker (campaign J) and unseen layouts (campaign L)

Campaign J ran Clef-Flash together with Qwen3.5-4B on 24 worlds, paired with v3.6: it fits on one 32 GB V100 (peak 28.0 GB) and behaved as expected in 23/24 (v3.6: 24/24, exact McNemar $p = 1.0000$), with unsafe deliveries 1 (v3.6: 0). The registered claim (two-model design on the simulated robot, no extra risk) is not supported; the one unsafe delivery was a small fire that both models missed, the correlated failure their development overlap predicts, so we did not adopt the pair (details: Supp. Sec. S11).

Unseen layouts (campaign L). Campaign L ran v3.6 once in each of 24 randomly drawn arena layouts with nominal, swapped and empty stations and small fires; it behaved as expected in **TBD** (95% CI **TBD**) with **TBD** unsafe deliveries. The registered claim that v3.6 generalises to unseen layouts is **TBD**.

VI. Discussion

Inputs, not just models, decide safety. Trusting memory is efficient but its risk grows with how often the world changes, which the robot cannot observe without looking (H). Observation 1, Corollary 1 and Proposition 2 make the question moot for safety: memory and any other advice decide the order and strictness of the checks, never their existence; the guarantee ends at the camera (Sec. V-F). **Moving beats thinking harder.** The largest safety gain came from moving 0.4 m (G), not from a larger model, a better prompt or a second model; a hidden fire carries no pixels to reason about. **What pre-registration bought us.** Four registered claims failed (T1's 2% bound, F, T2, J); the first two led to the look-around and J stopped us adopting a second model. Unregistered, each could have been rationalised away.

VII. Limitations

All results are from simulation: simple meshes, two supply types, fixed station geometry, one robot, a stylised fire, and sensor degradations rather than the appearance gap of real objects or smoke. Campaign H scripts its changes over three missions and two stations (V tests random changes). Corollary 1 and Proposition 2 assume a per-check false-approval bound uniform over the stations a route meets, and Proposition 2 protects only what enters through the advice interface: text in the scene is perception. Barely visible fires (21–613 pixels) are excluded, and the guard is hand-written, not synthesised [27], [28]. Threats to validity: Supp. Sec. S9.

VIII. Conclusion

Three lessons carry beyond our arena. An off-the-shelf local VLM can gate a physical action with few false approvals if it only scores closed-set answers and motion stays behind an independent guard (T1: 3 of 408; R: 0/2304). The failures that remained were failures of viewpoint, not of the model (G: 11 to 1 of 16). And memory should change *where* and *how carefully* a robot looks, never *whether* it looks (H: supported); the same holds for any advice it is given (A: **TBD**). A second, related model did not add safety (J: not supported). All of this is simulation; a site-specific threshold is the first thing a real deployment must calibrate. All code, protocols and per-trial evidence are released.

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